

5. Daemon Processes: When moving on to a rather special background process, Daemons start a system startup and run forever as a service. A user can control them via the init process. They are intentionally orphaned processes aimed to have a background process. Since they are created using a process forking a child process, their parent process is usually (not always) the init process. Interestingly, the daemon process performs operations such as dissociating it from any controlling terminal.

```
guest-mm2x7k@support21:~$ ps -axjf
PPID  PID  PGID  SID  TTY  TPGID  STAT  UID  TIME  COMMAND
0      2    0     0    0 ?    -1 S   0     0:00  \ [kthreadd]
2      3    0     0    0 ?    -1 I   0     0:00  \ [kworker/0:0]
2      4    0     0    0 ?    -1 I<  0     0:00  \ [kworker/0:0H]
2      5    0     0    0 ?    -1 I   0     0:00  \ [kworker/u8:0]
2      6    0     0    0 ?    -1 I<  0     0:00  \ [mm_percpu_wq]
2      7    0     0    0 ?    -1 S   0     0:00  \ [ksoftirqd/0]
2      8    0     0    0 ?    -1 I   0     0:01  \ [rcu_sched]
2      9    0     0    0 ?    -1 I   0     0:00  \ [rcu_bh]
2     10    0     0    0 ?    -1 S   0     0:00  \ [migration/0]
2     11    0     0    0 ?    -1 S   0     0:00  \ [watchdog/0]
2     12    0     0    0 ?    -1 S   0     0:00  \ [cpuhp/0]
2     13    0     0    0 ?    -1 S   0     0:00  \ [cpuhp/1]
2     14    0     0    0 ?    -1 S   0     0:00  \ [watchdog/1]
2     15    0     0    0 ?    -1 S   0     0:00  \ [migration/1]
2     16    0     0    0 ?    -1 S   0     0:00  \ [ksoftirqd/1]
2     18    0     0    0 ?    -1 I<  0     0:00  \ [kworker/1:0H]
2     19    0     0    0 ?    -1 S   0     0:00  \ [cpuhp/2]
2     20    0     0    0 ?    -1 S   0     0:00  \ [watchdog/2]
2     21    0     0    0 ?    -1 S   0     0:00  \ [migration/2]
2     22    0     0    0 ?    -1 S   0     0:00  \ [ksoftirqd/2]
2     24    0     0    0 ?    -1 I<  0     0:00  \ [kworker/2:0H]
2     25    0     0    0 ?    -1 S   0     0:00  \ [cpuhp/3]
```

Different types of processes in Linux

Running a Foreground Process

You can start a foreground process by either running it from the dashboard or the terminal. In case you're using the terminal, you need to wait until the foreground process is running. This particularly includes data-intensive tasks, which take a lot of time and processing power. It may also take hours to complete the process; hence, the programs are sent to the background and run accordingly.

Running a Background Process

To start with a background process from the terminal, start the program and press Ctrl+Z. It would help if you typed "bg" to send the process to the background. In cases where you want to bring the process back to the foreground, you can use the command "fg" and continue a program that was stopped.